CIE International A Level Physics

# Topic 25

# Astronomy and Cosmology

Revision Booklet

**This booklet covers:**

- Luminosity and Radiant Flux Intensity
- Standard Candles and Distance Measurement
- Wien's Displacement Law and Stellar Temperature
- The Stefan–Boltzmann Law and Stellar Radii
- Redshift and the Expanding Universe
- Hubble's Law and the Big Bang

## Luminosity and Radiant Flux Intensity

### Luminosity

The **luminosity** $L$ of a star is the total power of electromagnetic radiation emitted by the star in all directions.

$$L \quad \text{units: W (watts)}$$

Luminosity depends on the star's **surface temperature** and **surface area** — not on its distance from us.

### Radiant Flux Intensity

The **radiant flux intensity** $F$ (sometimes called apparent brightness) is the power of radiation received per unit area at a detector (e.g. a telescope on Earth).

$$F \quad \text{units: W m}^{-2}$$

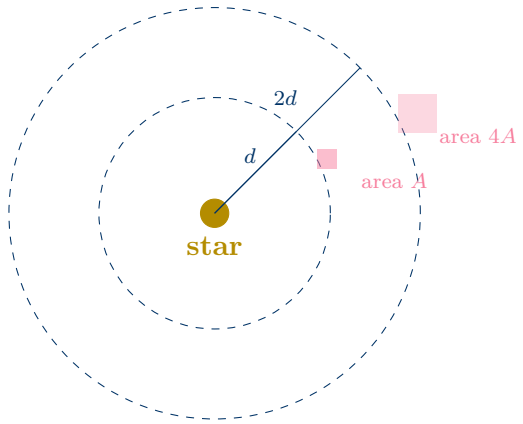
$F$ depends on both the luminosity of the source and its distance $d$.

### Inverse Square Law for Radiant Flux

Assuming the star radiates uniformly in all directions and there is no absorption by the intervening medium, the radiation spreads over a sphere of area $4\pi d^2$:

$$F = \frac{L}{4\pi d^2}$$

$F$ = radiant flux intensity at distance $d$ (W m$^{-2}$)
$L$ = luminosity of the star (W)
$d$ = distance from the star (m)




$F \propto 1/d^2$: double the distance, quarter the flux

## Standard Candles

### Standard Candle

A **standard candle** is an astronomical object whose **luminosity is known** (or can be determined independently of distance). By measuring the radiant flux intensity $F$ received from the object and using $F = L/(4\pi d^2)$, its distance $d$ can be calculated:

$$d = \sqrt{\frac{L}{4\pi F}}$$

### Standard Candles Used in Practice

- **Cepheid variable stars**: pulsating stars whose *period of variation* is related to their luminosity (period–luminosity relation). Measure the period $\Rightarrow$ know $L$ $\Rightarrow$ measure $F$ $\Rightarrow$ find $d$.
- **Type Ia supernovae**: thermonuclear explosions of white dwarf stars that all reach approximately the same peak luminosity. Visible across enormous distances (billions of light-years), making them useful for measuring distances to distant galaxies.

### Assumptions and Limitations

The inverse square law assumes:

- No absorption of radiation between source and detector (no dust or gas in the way).
- The source radiates **isotropically** (equally in all directions).

Dust absorption causes stars to appear **dimmer** than expected, which would lead to an **overestimate** of distance if uncorrected.

## Wien's Displacement Law and Stellar Temperature

### Black-Body Radiation

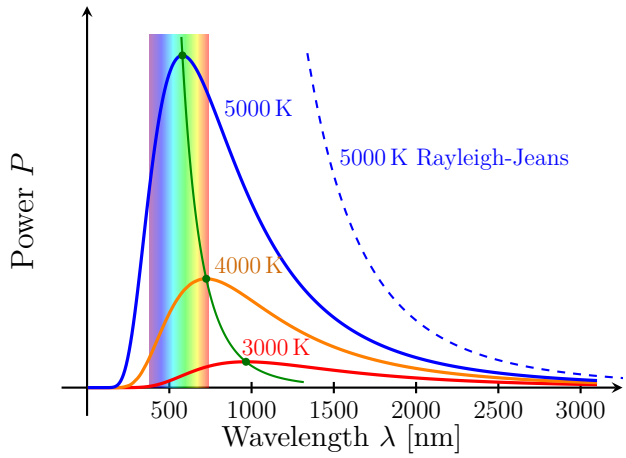
Stars approximate **black bodies** — objects that absorb all incident radiation and emit a characteristic continuous spectrum that depends only on temperature. The peak wavelength of this spectrum shifts with temperature.

### Wien's Displacement Law

$$\lambda_{\text{max}} \propto \frac{1}{T} \qquad \text{equivalently} \qquad \lambda_{\text{max}} T = b$$

$\lambda_{\text{max}}$ = wavelength at peak intensity of the spectrum (m)
$T$ = surface temperature of the star (K)
$b$ = Wien's constant = $2.90 \times 10^{-3}$ m K

A **hotter** star has its peak at a **shorter** wavelength (bluer colour). A **cooler** star peaks at a **longer** wavelength (redder colour).




## The Stefan–Boltzmann Law and Stellar Radii

### Stefan–Boltzmann Law

For a black-body sphere (approximating a star) of radius $r$ and surface temperature $T$:

$$L = 4\pi r^2 \sigma T^4$$

$L$ = luminosity (W)
$r$ = radius of the star (m)
$\sigma$ = Stefan–Boltzmann constant = $5.67 \times 10^{-8}$ W m$^{-2}$K$^{-4}$
$T$ = surface temperature (K)

### Estimating Stellar Radius

By combining Wien's law and the Stefan–Boltzmann law:

1. Measure $\lambda_{\text{max}}$ from the star's spectrum $\Rightarrow$ use Wien's law to find $T$.
2. Measure $F$ (radiant flux intensity) and find $d$ (e.g. via a standard candle or parallax).
3. Find $L$ from $L = 4\pi d^2 F$.
4. Rearrange Stefan–Boltzmann: $r = \sqrt{\dfrac{L}{4\pi\sigma T^4}}$

### Using Ratios

Exam questions often ask you to *compare* two stars rather than calculate absolute values. In that case, form a ratio to cancel constants:

$$\frac{L_1}{L_2} = \frac{r_1^2 T_1^4}{r_2^2 T_2^4}$$

This avoids large numbers and reduces the risk of errors.

## Redshift and the Expanding Universe

### Redshift

When a source of light moves **away** from an observer, the observed wavelength is **longer** (shifted towards the red end of the spectrum) than the wavelength emitted. This is the **Doppler effect** applied to light.
The lines in the emission or absorption spectrum of a distant galaxy are observed at **longer wavelengths** than those of the same element measured in the laboratory.

### Redshift Formula

For a source moving at speed $v \ll c$ relative to an observer:

$$z = \frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$

$z$ = redshift (dimensionless)
$\Delta\lambda$ = shift in wavelength (= observed $\lambda$ − emitted $\lambda$) (m)
$\lambda$ = emitted (rest-frame) wavelength (m)
$\Delta f$ = shift in frequency (Hz)
$v$ = recession speed of the source (m s$^{-1}$)

### Redshift as Evidence for Expansion

- Observations of distant galaxies show that their spectral lines are **all redshifted**.
- The greater the distance of a galaxy, the greater its redshift.
- This indicates that distant galaxies are moving **away** from us, and the further away they are, the faster they recede.
- This is consistent with the **Universe expanding**: it is space itself that is stretching, carrying galaxies apart, rather than the galaxies moving through space.

## Hubble's Law and the Big Bang Theory

### Hubble's Law

$$v \approx H_0\, d$$

$v$ = recession speed of a galaxy (m s$^{-1}$)
$H_0$ = Hubble constant (s$^{-1}$, though often quoted in km s$^{-1}$ Mpc$^{-1}$)
$d$ = distance of the galaxy from Earth (m)

In CIE examinations, SI units are used: $H_0$ in s$^{-1}$ and $d$ in metres.

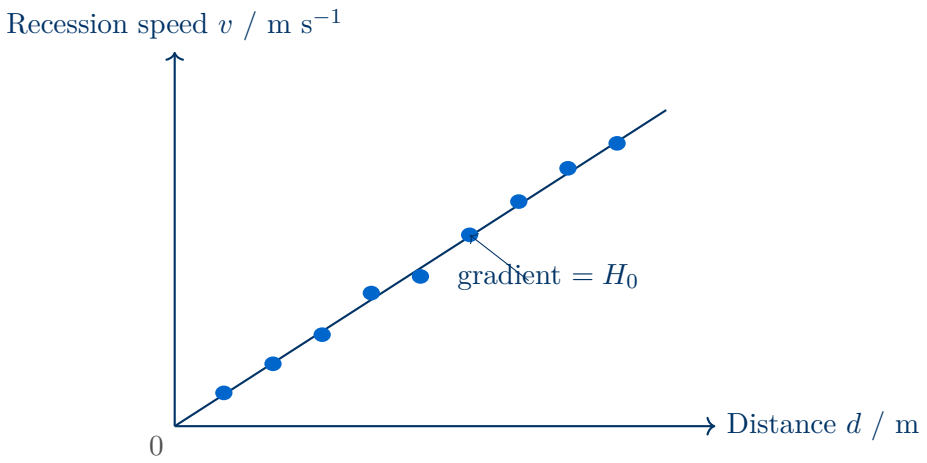
## Hubble's Law and the Big Bang Theory

- Hubble's law states that the recession speed of a galaxy is proportional to its distance.
- If all galaxies are currently moving apart, then in the past they must have been **closer together**.
- Extrapolating back in time, all matter was once concentrated in an extremely hot, dense state — the **Big Bang**.
- An estimate of the **age of the Universe** can be obtained from:

$$t \approx \frac{1}{H_0}$$

(This assumes a constant rate of expansion, which is a simplification.)

### Hubble Plot: Recession Speed vs Distance




## Determining $H_0$ from a Graph

A graph of recession speed $v$ against distance $d$ gives a straight line through the origin. The **gradient** of this line is the Hubble constant $H_0$. In practice, there is significant scatter due to the difficulty of measuring distances to distant galaxies accurately.

## Formula Summary Sheet

| Formula | Quantity | Units |
|---|---|---|
| $F = L/(4\pi d^2)$ | Inverse square law / flux | W m$^{-2}$ |
| $\lambda_{\text{max}} T = b$ | Wien's displacement law | m K |
| $L = 4\pi r^2 \sigma T^4$ | Stefan–Boltzmann law | W |
| $\Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ | Redshift formula | — |
| $v \approx H_0 d$ | Hubble's law | m s$^{-1}$, m |
| $t \approx 1/H_0$ | Age of Universe estimate | s |

**Constants:** $\sigma = 5.67 \times 10^{-8}$ W m$^{-2}$K$^{-4}$, $b = 2.90 \times 10^{-3}$ m K, $c = 3.00 \times 10^{8}$ m s$^{-1}$

## Exam Technique and Problem-Solving Strategy

### Step-by-Step for Stellar Radius Problems

1. Find $T$: measure $\lambda_{\text{max}}$ from spectrum $\Rightarrow T = b/\lambda_{\text{max}}$.
2. Find $L$: use standard candle or parallax to get $d$; then $L = 4\pi d^2 F$.
3. Find $r$: rearrange Stefan–Boltzmann: $r = \sqrt{L/(4\pi\sigma T^4)}$.

### Common Errors — Avoid These!

- Forgetting the $4\pi$ in both $F = L/4\pi d^2$ and $L = 4\pi r^2 \sigma T^4$.
- Confusing **luminosity** (intrinsic power, independent of distance) with **flux intensity** (observed brightness, depends on distance).
- Using $\Delta\lambda = \lambda_{\text{obs}} - \lambda_{\text{emitted}}$ but getting the **sign wrong**: for a receding source, $\Delta\lambda > 0$ (observed wavelength is longer).
- Applying $v = H_0 d$ with $d$ in Mpc and $H_0$ in km s$^{-1}$ Mpc$^{-1}$ — in CIE exams **always convert to SI** (metres and s$^{-1}$).
- Stating that the Big Bang means galaxies are moving **through space** — more accurately, **space itself is expanding**.

## Worked Examples

### Example 1 — Stellar Radius from Wien and Stefan–Boltzmann

**Question:** A star has a peak emission wavelength of 480 nm and a luminosity of $5.2 \times 10^{26}$ W. Estimate its radius. ($b = 2.90 \times 10^{-3}$ m K, $\sigma = 5.67 \times 10^{-8}$ W m$^{-2}$K$^{-4}$)

**Solution**

**Solution:**
Step 1 — Surface temperature:

$$T = \frac{b}{\lambda_{\text{max}}} = \frac{2.90 \times 10^{-3}}{480 \times 10^{-9}} = 6040 \text{ K}$$

Step 2 — Rearrange Stefan–Boltzmann:

$$r = \sqrt{\frac{L}{4\pi\sigma T^4}} = \sqrt{\frac{5.2 \times 10^{26}}{4\pi \times 5.67 \times 10^{-8} \times (6040)^4}}$$

$$T^4 = (6.04 \times 10^3)^4 = 1.33 \times 10^{15} \text{ K}^4$$

$$r = \sqrt{\frac{5.2 \times 10^{26}}{4\pi \times 5.67 \times 10^{-8} \times 1.33 \times 10^{15}}} = \sqrt{\frac{5.2 \times 10^{26}}{9.47 \times 10^8}} = \sqrt{5.49 \times 10^{17}} = \mathbf{7.4 \times 10^8} \text{ m}$$

### Example 2 — Distance Using Flux and Luminosity

**Question:** A type Ia supernova has a peak luminosity of $2.0 \times 10^{36}$ W. It is observed with a radiant flux intensity of $3.5 \times 10^{-14}$ W m$^{-2}$. Calculate its distance.

**Solution**

**Solution:**

$$d = \sqrt{\frac{L}{4\pi F}} = \sqrt{\frac{2.0 \times 10^{36}}{4\pi \times 3.5 \times 10^{-14}}}$$

$$d = \sqrt{\frac{2.0 \times 10^{36}}{4.40 \times 10^{-13}}} = \sqrt{4.55 \times 10^{48}} = \mathbf{2.1 \times 10^{24}} \text{ m}$$

### Example 3 — Recession Speed and Age of Universe

**Question:** A galaxy shows a spectral line at 656 nm that is observed at 689 nm. The Hubble constant is $H_0 = 2.2 \times 10^{-18}$ s$^{-1}$. Calculate (a) the recession speed of the galaxy, (b) its distance, and (c) an estimate of the age of the Universe.

**Solution**

**Solution:**
(a) $\dfrac{\Delta\lambda}{\lambda} = \dfrac{689 - 656}{656} = \dfrac{33}{656} = 0.0503$
$v = 0.0503 \times c = 0.0503 \times 3.00 \times 10^8 = \mathbf{1.51 \times 10^7}$ m s$^{-1}$

(b) $d = v/H_0 = 1.51 \times 10^7 / 2.2 \times 10^{-18} = \mathbf{6.9 \times 10^{24}}$ m
(c) $t \approx 1/H_0 = 1/(2.2 \times 10^{-18}) = \mathbf{4.5 \times 10^{17}}$ s ($\approx$ 14 billion years)

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** Define (a) luminosity and (b) radiant flux intensity. State the relationship between them and the distance $d$ to the source.

*[4 marks]*

**Q2.** The Sun has a surface temperature of 5800 K. Calculate the peak wavelength of its emission spectrum and state what colour this corresponds to.

*[2 marks]*

**Q3.** Explain what is meant by a *standard candle* and describe one example of a standard candle used in astronomy.

*[3 marks]*

**Q4.** A hydrogen spectral line has a rest wavelength of 434 nm. In the spectrum of a distant galaxy it is observed at 461 nm. Calculate the recession speed of the galaxy.

*[3 marks]*

**Q5.** Explain how observations of redshift from distant galaxies lead to the conclusion that the Universe is expanding.

*[3 marks]*

## Section B — Longer Structured Questions

**Q6.** Star A has surface temperature 12 000 K and radius $3.5 \times 10^9$ m. Star B has the same luminosity as Star A but a surface temperature of 4500 K.

(a) Calculate the luminosity of Star A.

*[2 marks]*

(b) Calculate the radius of Star B.

*[3 marks]*

(c) State which star would appear bluer and explain why.

*[2 marks]*

**Q7.** A galaxy is observed to have a recession speed of $4.8 \times 10^6$ m s$^{-1}$. The Hubble constant is $H_0 = 2.2 \times 10^{-18}$ s$^{-1}$.

(a) Calculate the distance of the galaxy.

*[2 marks]*

(b) A spectral line in the galaxy's spectrum has an emitted wavelength of 589 nm. Calculate the observed wavelength.

*[2 marks]*

(c) Explain how Hubble's law provides evidence for the Big Bang theory.

*[3 marks]*

## Mark Scheme and Answers

**Q1.** (a) Luminosity: the total power of electromagnetic radiation emitted by a star (in all directions) [2]. (b) Radiant flux intensity: the power of radiation received per unit area at the observer's location [1]; $F = L/(4\pi d^2)$ [1].

**Q2.** $\lambda_{\text{max}} = b/T = 2.90 \times 10^{-3}/5800 = \mathbf{500}$ nm [1]; this corresponds to green light (near the centre of the visible spectrum) [1].

**Q3.** A standard candle is an astronomical object of known luminosity [1]; example: type Ia supernova (all reach the same peak luminosity) [1]; or Cepheid variable (period of brightness variation gives luminosity via period–luminosity relation) [1].

**Q4.** $\Delta\lambda = 461 - 434 = 27$ nm [1]; $v = c\Delta\lambda/\lambda = 3.00 \times 10^8 \times 27/434$ [1] $= \mathbf{1.87 \times 10^7}$ m s$^{-1}$ [1].

**Q5.** Spectral lines from distant galaxies are observed at longer wavelengths (redshifted) than expected [1]; this indicates the galaxies are moving away from us [1]; the greater the

distance, the greater the redshift/recession speed — consistent with all space expanding uniformly [1].

**Q6(a).** $L = 4\pi r^2 \sigma T^4 = 4\pi \times (3.5 \times 10^9)^2 \times 5.67 \times 10^{-8} \times (12000)^4$ [1]; $= 4\pi \times 1.225 \times 10^{19} \times 5.67 \times 10^{-8} \times 2.07 \times 10^{16} = \mathbf{1.80 \times 10^{29}}$ W [1].

**Q6(b).** Same $L$; $r_B^2 T_B^4 = r_A^2 T_A^4$ [1]; $r_B = r_A (T_A/T_B)^2 = 3.5 \times 10^9 \times (12000/4500)^2 = 3.5 \times 10^9 \times 7.11$ [1] $= \mathbf{2.49 \times 10^{10}}$ m [1].

**Q6(c).** Star A is bluer [1]; it has a higher surface temperature, so by Wien's law its peak wavelength is shorter (towards the blue end of the spectrum) [1].

**Q7(a).** $d = v/H_0 = 4.8 \times 10^6 / 2.2 \times 10^{-18} = \mathbf{2.18 \times 10^{24}}$ m [2].

**Q7(b).** $\Delta\lambda = \lambda v/c = 589 \times 10^{-9} \times 4.8 \times 10^6 / 3.00 \times 10^8 = 9.42 \times 10^{-9}$ m [1]; observed $\lambda = 589 + 9.4 = \mathbf{598}$ nm [1].

**Q7(c).** Hubble's law shows recession speed is proportional to distance — the further away a galaxy, the faster it recedes [1]; this means all galaxies are moving apart from one another, implying the Universe is expanding [1]; extrapolating back in time, all matter must have originated from a single point — the Big Bang [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Define luminosity and radiant flux intensity | |
| ☐ | Use the inverse square law $F = L/(4\pi d^2)$ | |
| ☐ | Define a standard candle and explain how it is used to find distance | |
| ☐ | Give examples of standard candles (Cepheid variables, Type Ia supernovae) | |
| ☐ | State and apply Wien's displacement law $\lambda_{\text{max}}T = b$ | |
| ☐ | State and apply the Stefan–Boltzmann law $L = 4\pi r^2 \sigma T^4$ | |
| ☐ | Combine Wien and Stefan–Boltzmann to estimate stellar radius | |
| ☐ | Explain what redshift is and how it is observed in galaxy spectra | |
| ☐ | Use $\Delta\lambda/\lambda \approx v/c$ to find recession speeds | |
| ☐ | Explain how redshift provides evidence for an expanding Universe | |
| ☐ | State and apply Hubble's law $v \approx H_0 d$ (SI units) | |
| ☐ | Use $t \approx 1/H_0$ as an estimate for the age of the Universe | |
| ☐ | Explain how Hubble's law leads to the Big Bang theory | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

Cosmology asks the biggest questions in physics — and answers them with the same tools you've used all year. A few formulas, careful unit conversions, and the ability to interpret a graph are all you need to explore the scale of the Universe itself.